

Differences between How Mathematics is Taught in Saint Lucia and in Japan

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Generally speaking, as the college students in Saint Lucia have scientific calculators, concrete phenomena in a real world seem to be more focused on than in Japan.

On the contrary, as the high school students in Japan are not allowed to have any calculators during the sessions, abstract mathematical concepts are more focused on than in Saint Lucia. We, Japanese teachers, think that without understanding abstract mathematical concepts well the students couldn't apply those to the real world problems.

I think it is the greatest difference between how mathematics is taught in Saint Lucia and in Japan. Could you please let me share the concrete examples as of now down below with you?

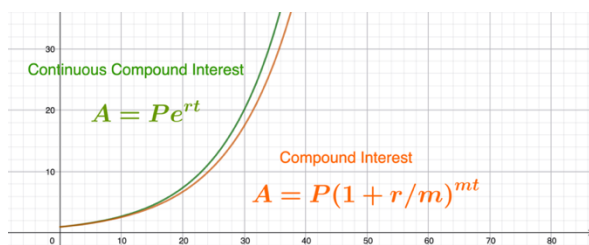
Consumer Arithmetic

In Japan we never teach the material of Consumer Arithmetic. Now I think even in Japan we should teach the consumer arithmetic, especially a hire purchase and a depreciation schedule.

If we taught the materials of Consumer Arithmetic in Japan, we wouldn't use the formulas without checking the proof of them.

1. Compounded Continuously

$$\boxed{\text{Future Value} = Pe^{rt}} \quad \Leftrightarrow \quad F = \lim_{n \rightarrow \infty} P\left(1 + \frac{r}{n}\right)^{nt} = \lim_{\frac{n}{r} \rightarrow \infty} P\left(1 + \frac{r}{n}\right)^{\frac{n}{r} \times rt} = Pe^{rt}$$



For Hands-on Experience



<https://www.geogebra.org/classic/wafjghyd>

2. Ordinary Annuities

$$\boxed{\text{Future Value} = PMT \left(\frac{(1 + \frac{r}{n})^{nt} - 1}{\frac{r}{n}} \right)}$$



$$\text{Future Value} = \overset{\text{1st Payment}}{PMT(1 + \frac{r}{n})^{nt-1}} + \overset{\text{2nd Payment}}{PMT(1 + \frac{r}{n})^{nt-2}} + \cdots + \overset{\text{(nt-1)th Payment}}{PMT(1 + \frac{r}{n})^1} + \overset{\text{Last Payment}}{PMT}$$

$$= PMT \left(\frac{(1 + \frac{r}{n})^{nt} - 1}{(1 + \frac{r}{n}) - 1} \right) \quad \text{Sum of a Geometric Sequence}$$

$$= PMT \left(\frac{(1 + \frac{r}{n})^{nt} - 1}{\frac{r}{n}} \right)$$

$$\boxed{\text{Present Value} = PMT \left(\frac{1 - (1 + \frac{r}{n})^{-nt}}{\frac{r}{n}} \right)}$$



$$\text{Present Value} \left(1 + \frac{r}{n} \right)^{nt} = PMT \left(\frac{(1 + \frac{r}{n})^{nt} - 1}{\frac{r}{n}} \right)$$

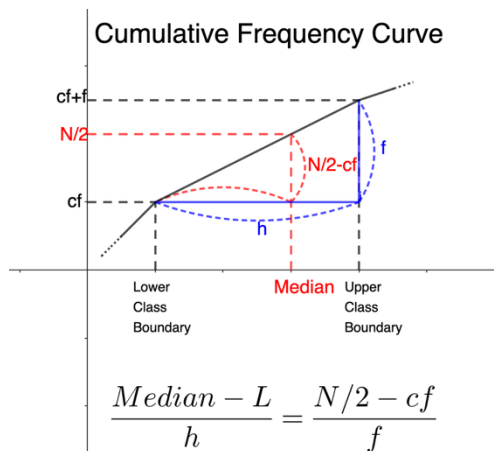
$$\text{Present Value} = \frac{PMT \left(\frac{(1 + \frac{r}{n})^{nt} - 1}{\frac{r}{n}} \right)}{(1 + \frac{r}{n})^{nt}} = PMT \left(\frac{1 - (1 + \frac{r}{n})^{-nt}}{\frac{r}{n}} \right)$$

Statistics

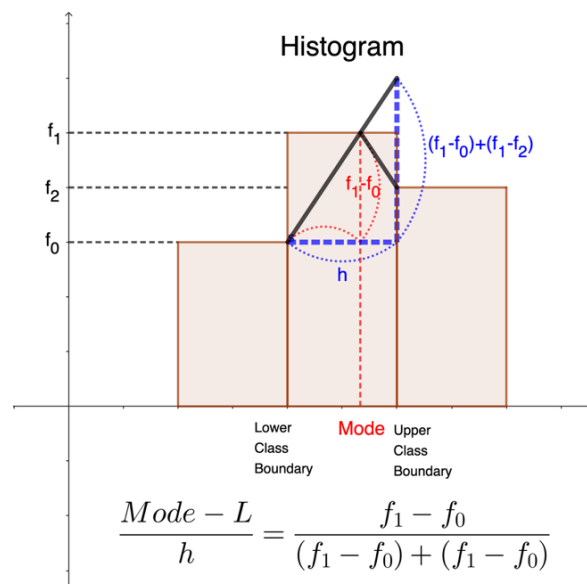
In Japan we teach Relative Frequency Polygon at the same time as Frequency Polygon so that we can compare the different distributions with each other like Pie Chart.

In Japan we never teach Median nor Mode of Grouped Data. If we taught them in Japan, we wouldn't use the formulas without checking the proof of them.

$$\boxed{\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h}$$

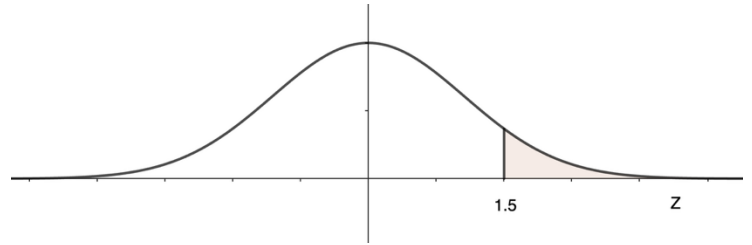


$$\boxed{\text{Mode} = L + \left(\frac{f_1 - f_0}{(f_1 - f_0) + (f_1 - f_2)} \right) \times h}$$

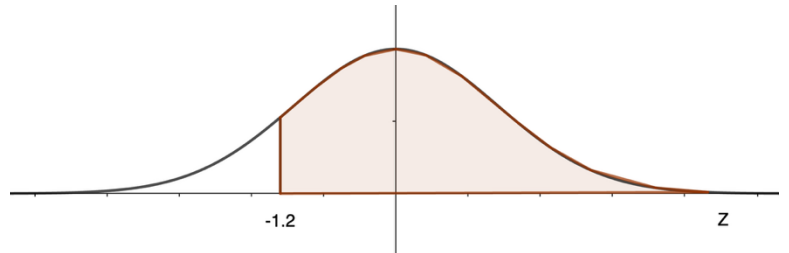


In Japan whenever we solve the probability problem on the standard normal distribution, we always let our students draw the picture of the distribution so that they can understand the relations like:

$$P(Z > 1.5) = 1 - P(Z < 1.5)$$

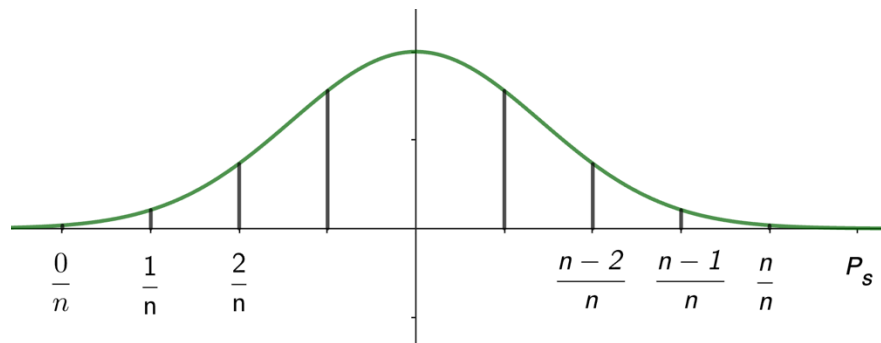


$$P(Z > -1.2) = P(Z < 1.2)$$

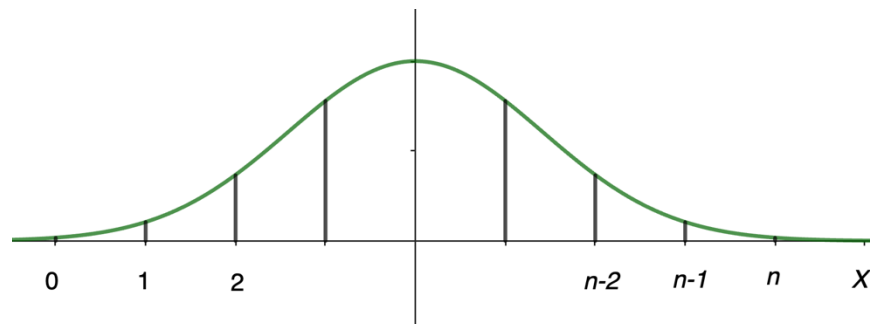


In Japan we may refer to the relation between the Binomial Distribution and the Distribution of the Sample Proportion so that the students can understand well the concept of the continuity correction.

The Binomial Distribution $B(n, p) \sim N(np, np(1 - p))$



The Distribution of the Sample Proportion $P_s \sim N(p, \frac{p(1-p)}{n})$



In Japan whenever we solve the problem of the confidence interval of the population mean, we let the students draw a picture so that they can understand the meaning of the formulas.

1. 90% Confidence Interval for the Population Mean

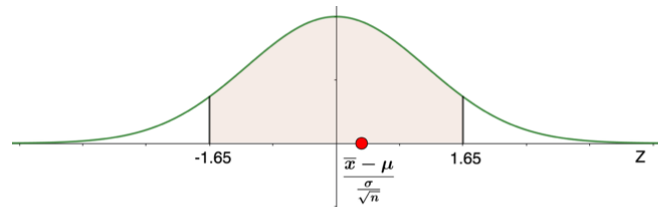
Case 1: $\bar{x} - 1.65 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.65 \frac{\sigma}{\sqrt{n}}$ Population: normal, σ : known



$$-1.65 \leq \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq 1.65$$

Hence $-1.65 \frac{\sigma}{\sqrt{n}} \leq \bar{x} - \mu \leq 1.65 \frac{\sigma}{\sqrt{n}}$

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$



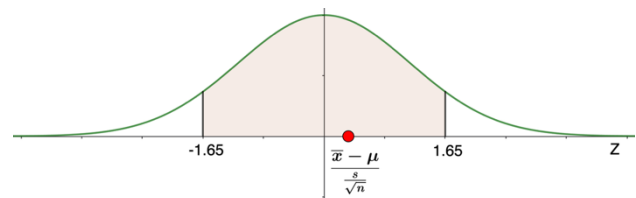
Case 2: $\bar{x} - 1.65 \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.65 \frac{s}{\sqrt{n}}$ Population: not normal, σ : unknown, n : large



$$-1.65 \leq \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \leq 1.65$$

Hence $-1.65 \frac{s}{\sqrt{n}} \leq \bar{x} - \mu \leq 1.65 \frac{s}{\sqrt{n}}$

$$\frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \sim N(0, 1)$$



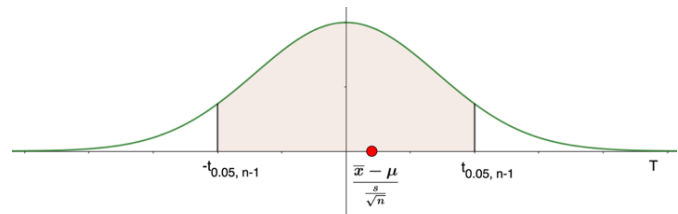
Case 3: $\bar{x} - t_{0.05, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{0.05, n-1} \frac{s}{\sqrt{n}}$ Population: not normal, σ : unknown, n : small



$$-t_{0.05, n-1} \leq \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \leq t_{0.05, n-1}$$

Hence $-t_{0.05, n-1} \frac{s}{\sqrt{n}} \leq \bar{x} - \mu \leq t_{0.05, n-1} \frac{s}{\sqrt{n}}$

$$\frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \sim T_{df=n-1}$$



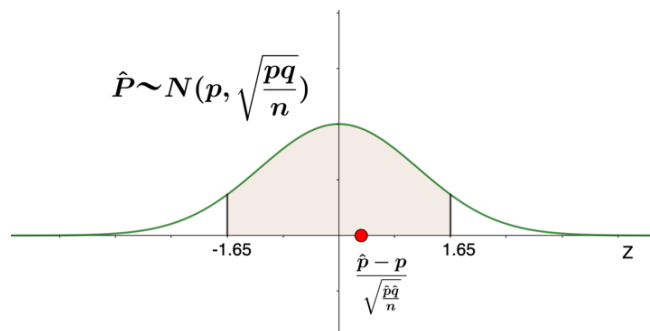
2. 90% Confidence Interval for the Population Proportion

$$\hat{p} - 1.65\sqrt{\frac{\hat{p}\hat{q}}{n}} \leq p \leq \hat{p} + 1.65\sqrt{\frac{\hat{p}\hat{q}}{n}}$$



$$-1.65 \leq \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}\hat{q}}{n}}} \leq 1.65$$

Hence $-1.65\sqrt{\frac{\hat{p}\hat{q}}{n}} \leq \hat{p} - p \leq 1.65\sqrt{\frac{\hat{p}\hat{q}}{n}}$

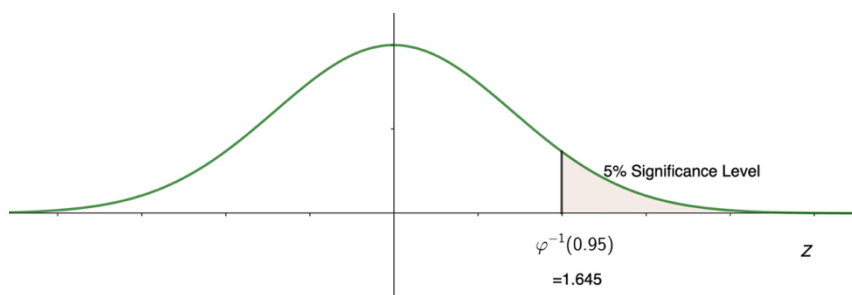


In Japan whenever we solve the problem of the Hypothesis Testing of the Population Mean, we let the students draw a picture so that the students easily tell the difference between the Z-value and the probability.

Null Hypothesis $H_0 : \mu = \mu_0$

Alternative Hypothesis $H_1 : \mu > \mu_0$

Test Statistics $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$



Trigonometry

In Japan when we define the trigonometry of the angle of more than 90 degree, we don't use the reference angle and the corresponding sign of the quadrant. But we use the coordinates of the point on the unit circle.

reference angle = 30°

$$\cos 30^\circ = \frac{1}{2} \quad \sin 30^\circ = \frac{\sqrt{3}}{2}$$

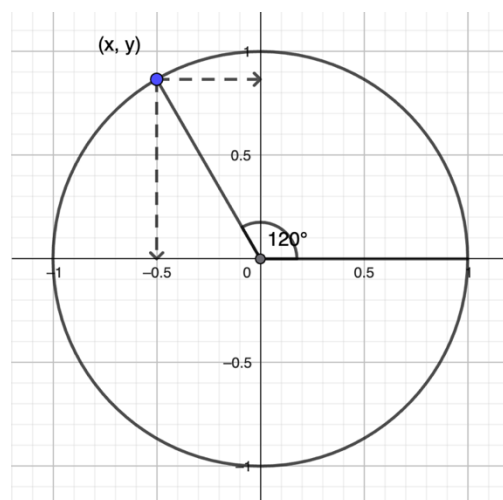
The angle of 120° is in the 2nd quadrant.

So, the cosine is negative and the sine is positive

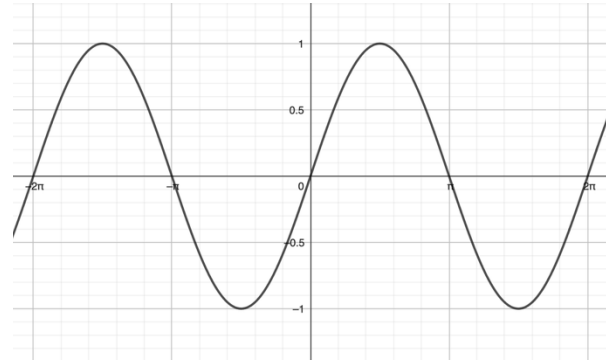
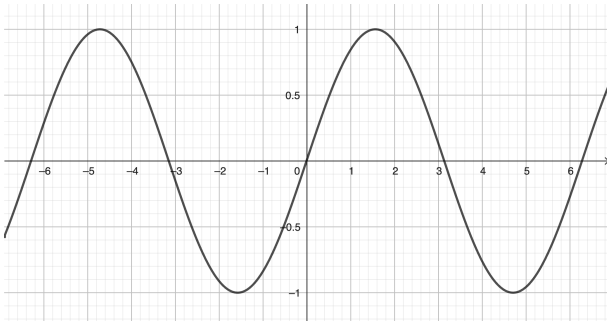
Hence, $\cos 120^\circ = -\frac{1}{2} \quad \sin 120^\circ = \frac{\sqrt{3}}{2}$



$$\cos 120^\circ = x = -\frac{1}{2} \quad \sin 120^\circ = y = \frac{\sqrt{3}}{2}$$



In Japan when we draw a sinusoidal wave, we never have x-axis with a mere radius-unit but have one with π -unit as radian.



In Japan when we refer to De Moivre's Theorem, we may refer to the additional theorem of the trigonometry.

$$(\cos\theta + i\sin\theta)^2 = \cos 2\theta + i\sin 2\theta$$



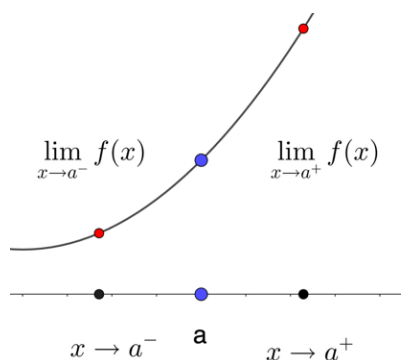
$$(\text{left hand side}) = \cos^2\theta + 2i\sin\theta\cos\theta + i^2\sin^2\theta$$

$$(\text{right hand}) = \cos 2\theta + i\sin 2\theta = \cos^2\theta - \sin^2\theta + i * 2\sin\theta\cos\theta$$

Calculus

As the students in St. Lucia have their own calculators, they calculate the output y-values to recognize to which value they are approaching when the input x-values are approaching to a certain value.

On the contrary, as the students in Japan have no calculators, they just recognize the situations by drawing the graphs.



For Hands-on Experience

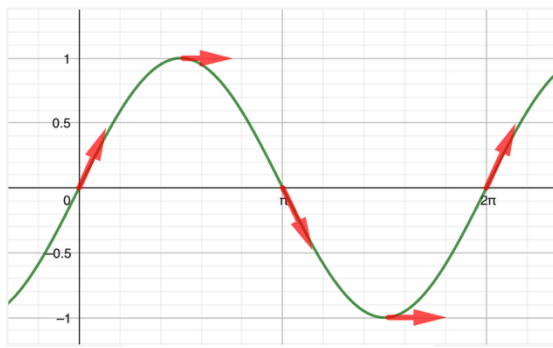


<https://www.geogebra.org/classic/pqfjdt8c>

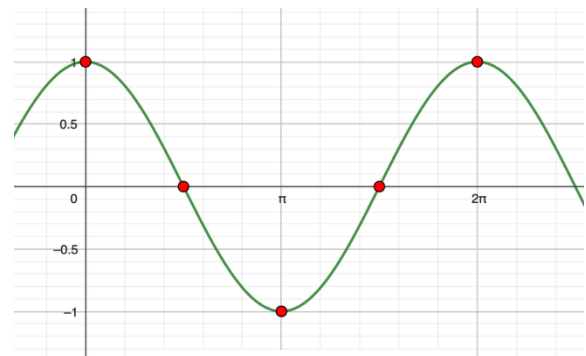
In Japan when we solve the indeterminate form, we don't compare just the driving forces but transform the whole expression into the determinate form.

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 5x + 4}{2x^2 + x + 3} = \lim_{x \rightarrow \infty} \frac{3 + \frac{5}{x} + \frac{4}{x^2}}{2 + \frac{1}{x} + \frac{3}{x^2}} = \frac{3 + 0 + 0}{2 + 0 + 0} = \frac{3}{2}$$

In Japan, for example, when we teach the formula $(\sin x)' = \cos x$, we teach the meaning of this formula, too.



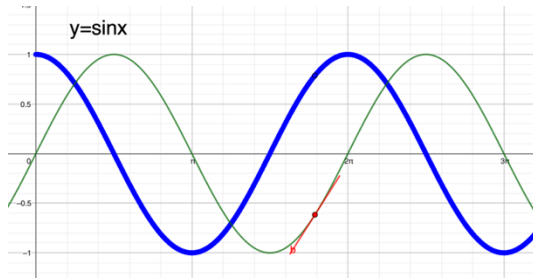
$$y = \sin x$$



$$y' = \cos x$$

The students in St. Lucia are likely just to remember the formula, for example, $(\sin x)' = \cos x$.

On the contrary, in Japan the graph of the function helps the students to understand how the gradient function looks.



For Hands-on Experience



<https://www.geogebra.org/classic/mqjahg8y>

In Japan when we investigate the nature of the stationary point, we don't check the sign of the second derivative if we can draw the graph of the first derivative.

The given function $y = x^3 - 3x^2 + 1$

$$y' = 3x(x - 2)$$

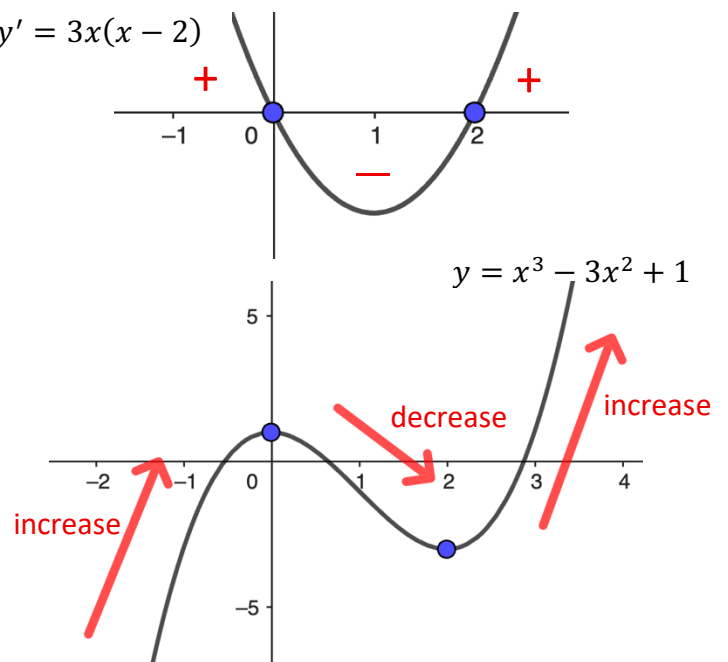
The first derivative $y' = 3x^2 - 6x = 3x(x - 2)$

Hence, the stationary points are $(0, 1)$ and $(2, -3)$

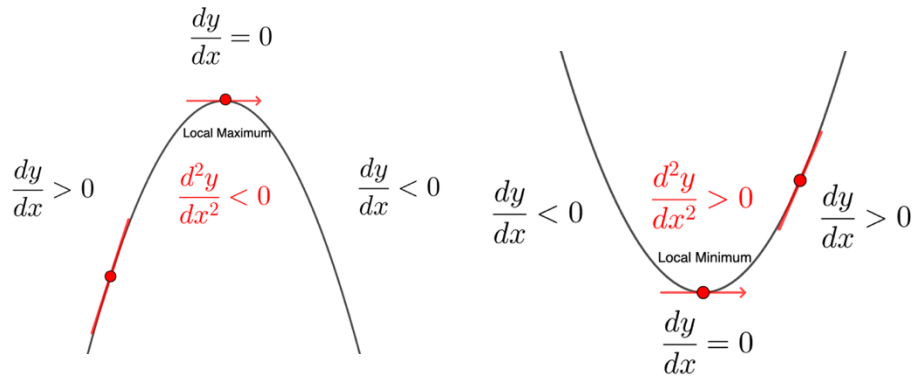


Hence, the stationary point $(0, 1)$ is Maximum.

And the stationary point $(2, -3)$ is minimum.



The students in St. Lucia check the second derivative to know the nature of the stationary points. On the contrary, the students in Japan draw the graph of the differentiated function to decide the nature of the stationary points by the changing signs of it.



For Hands-on Experience



<https://www.geogebra.org/classic/reptgxz>



<https://www.geogebra.org/classic/nkrsmrkj>

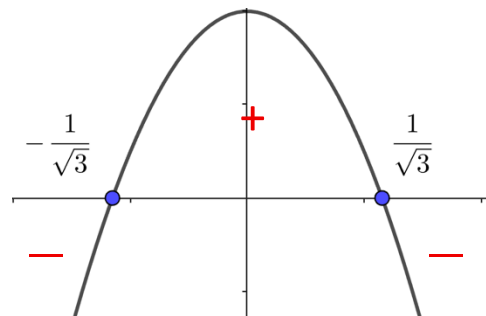
In Japan when we find the points of the inflection, we don't check the sign of any specific point if we can draw the graph of the second derivative.

The given function $y = 2x^2 - x^4 = x^2(\sqrt{2} + x)(\sqrt{2} - x)$

The first derivative $y' = 4x - 4x^3 = 4x(1 + x)(1 - x)$

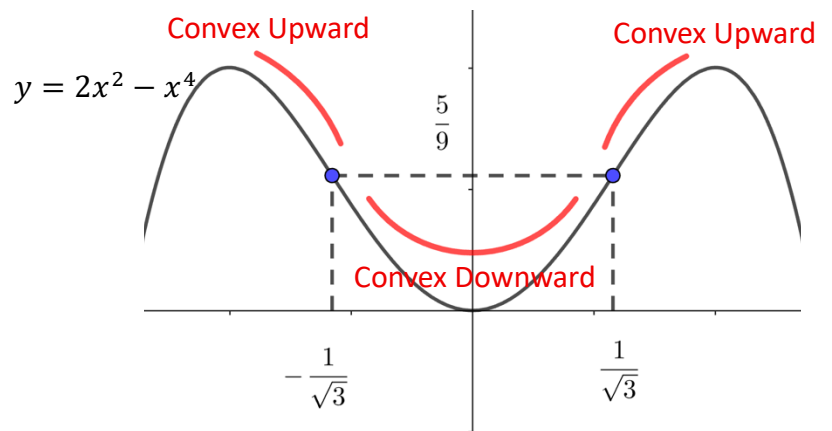
The second derivative $y'' = 4 - 12x^2 = 4(1 + \sqrt{3}x)(1 - \sqrt{3}x)$

$$y'' = 4(1 + \sqrt{3}x)(1 - \sqrt{3}x)$$



Hence, the points of inflection are

$$\left(-\frac{1}{\sqrt{3}}, \frac{5}{9}\right) \quad \text{and} \quad \left(\frac{1}{\sqrt{3}}, \frac{5}{9}\right)$$



In Japan when we use the Newton-Raphson method, we usually teach how to make the recurrence formula at the same time.

The tangent going through the point of $(x_n, f(x_n))$

$$y - f(x_n) = f'(x_n)(x - x_n)$$

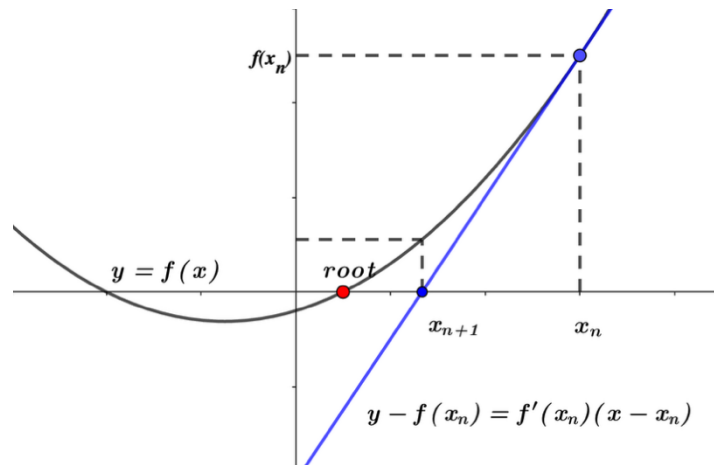
crosses the x-axis at the point of $(x_{n+1}, 0)$.

So, we can substitute $x = x_{n+1}, y = 0$

into the tangent expression.

$$0 - f(x_n) = f'(x_n)(x_{n+1} - x_n)$$

Hence
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



In Japan, the derivatives of trigonometric functions, exponential functions and logarithmic functions are all necessarily expected to be proved.

In Japan, we also need to prove the formulas of the product rule and the quotient rule.

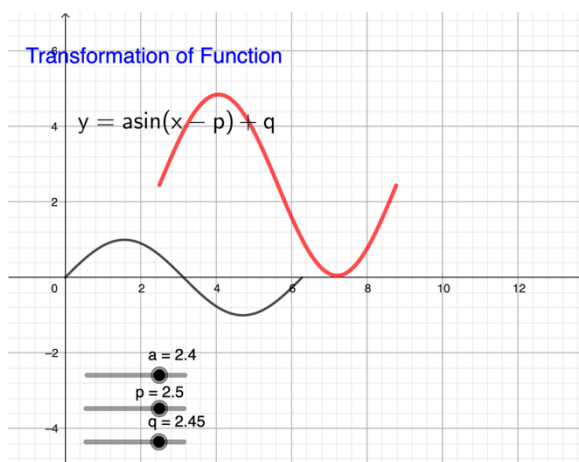
Algebra

In Japan we do not use the notation “ln” for the log natural. Instead, we just hide the base of “e” like “logP.”

In Japan we do not use the “ac-b method”, when we factorize the quadratic equations. Instead, we think about the suitable combination of 2 factors of the coefficient “a” and corresponding 2 factors of the other coefficient “c” in trials & errors method.

A transformation of a function is realized just on several points here in St. Lucia.

On the contrary, in Japan the students draw the result graph to show the transformation of the function.

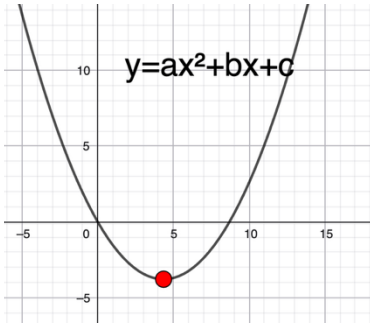


For Hands-on Experience



<https://www.geogebra.org/classic/fs2ev6tx>

In St. Lucia, to get the vertex of the quadratic function the students are required to remember the formula. On the contrary, in Japan, to get the vertex of the quadratic function the students are required to calculate.

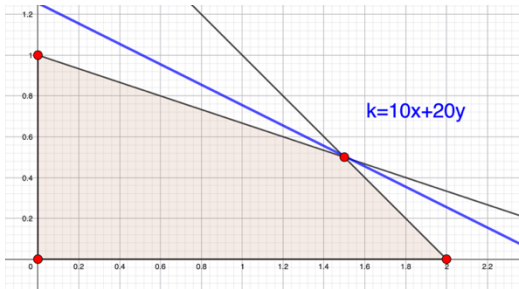


For Hands-on Experience



<https://www.geogebra.org/classic/ymjzt4ns>

To think the optimal situation with the linear programming method, the Lucian students investigate the specific points within the constraint area. On the contrary, in Japan the students need to draw the line to show which point represents the optimal situation.



For Hands-on Experience



<https://www.geogebra.org/classic/nfdxks9y>

In Japan we seldom draw lines of best fit or curves of best fit.

For Hands-on-Experience

Title	Line of Best Fit	Curve of Best Fit
QR code		
URL	https://www.geogebra.org/classic/r6xhx7sz	https://www.geogebra.org/classic/cagpp6jc

In Japan we seldom apply the mathematical concept to the real business model. So, the materials in Saint Lucia are sometimes eye-opening opportunities for me.

